Temporal Neuron Pruning for Energy-Efficient Spiking Neural Networks

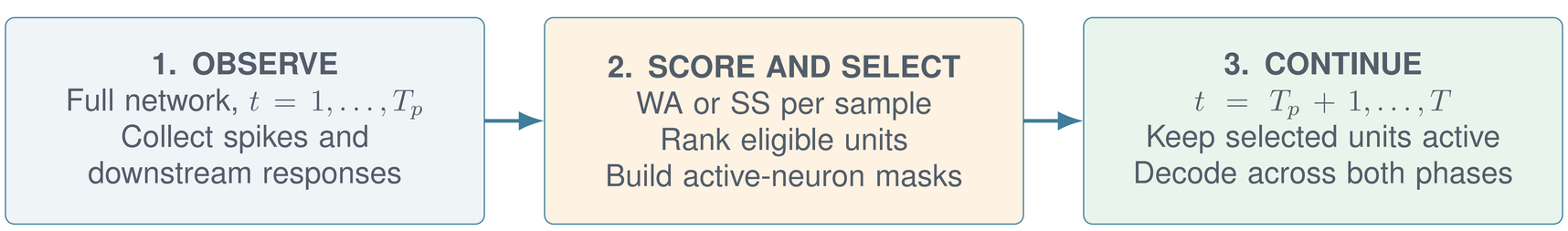
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SUMMARY OF RESEARCH

Motivation. Spike sparsity alone does not prevent redundant membrane updates and downstream computation. Fixed pruning cannot adapt the active neuron set to each input.

Goal. Use early spike activity to select neurons or channels during inference, preserving classification accuracy while reducing the remaining computation. Evaluate this trade-off across FC, CNN, and STDP-trained SNNs.



Pruning decisions are made once per sample. The main accuracy metric uses all timesteps (All readout); Phase-2-only accuracy is evaluated separately.

METHOD

Two-phase inference

- Observe for $T_p = \max(1, \lfloor \rho T \rfloor)$ timesteps; collect spike counts and downstream peak membrane potentials.
- Rank units by WA or SS, retain a target count, and keep that active set fixed during Phase 2.
- Earlier pruning leaves more time for savings but less evidence for ranking.

Weight-activity (WA) and sample-specific (SS)

$$s_l^{\mathrm{WA}} = c_l \odot \|W_{l+1}\|_{2,\mathrm{col}},$$
$$s_l^{\mathrm{SS}} = c_l \odot W_{l+1}[\mathrm{winner}, :].$$

c_l: Phase-1 spike count. W_{l+1}: outgoing weights restricted to retained downstream neurons. The winner has the largest downstream Phase-1 peak membrane potential.

- WA combines activity with outgoing connection strength.
- FC SS uses signed weights feeding the provisional winner; CNN SS uses spatial kernel magnitudes.

Reverse cascade for multiple FC hidden layers Start from the unpruned output layer and select hidden layers backward, $L \to \cdots \to 1$. Each score credits only surviving downstream neurons.

EVALUATION SETUP

Dataset / network	T	Test images
MNIST MLP, 784:100:10	20	5,000
MNIST MLP, 784:100:100:10	20	5,000
CIFAR-10 Spiking ResNet-18	6	10,000
MNIST STDP, 784:500	250	10,000

Environment. Python evaluation; deterministic MLP inference with constant input current. STDP uses Poisson spike trains (up to 128 Hz, $\Delta t = 1$ ms).

- **FC:** prune hidden neurons; keep output neurons. All readout uses a no-reset run.
- **CNN:** prune only intra-block middle channels. Keep shortcuts and block outputs; recalibrate BNTT offline for each criterion, timing, and ratio [1].
- **STDP:** score excitatory neurons using the class-proportion matrix; retain state and clamp pruned neurons below threshold [2,3].

The unpruned baseline uses all T timesteps. Pruning ratios refer to the eligible neurons or channels, rather than the whole network.

Activity-based clock-gated LIF model

$$E_{\mathrm{neuron}} = \sum_l [E_{\mathrm{spike}} S_l + E_{\mathrm{idle}} \{H_l T_p + K_l (T - T_p)\}] .$$

S_l: test-image-averaged spikes over both phases; H_l: full layer size; K_l: retained neurons. Post-synthesis flow: Design Compiler/ModelSim/PrimeTime; FREEPDK45, 1.1 V, typical corner, 100 MHz: $E_{\mathrm{spike}} = 0.4281$ pJ and $E_{\mathrm{idle}} = 0.4977$ pJ per neuron per timestep. Energy is a neuron-component estimate, excluding selection overhead and synaptic-memory access.

RESULTS

MLP: 90% hidden-neuron pruning after 4/20 timesteps

	1 hidden layer	2 hidden layers
Baseline accuracy	95.92%	96.26%
Baseline est. neuron energy	1.42 nJ	2.81 nJ
WA accuracy (All)	91.64%	92.86%
SS accuracy (All)	95.06%	95.38%
SS accuracy loss	0.86 pp	0.88 pp
WA est. energy reduction	63.9%	68.4%
SS est. energy reduction	**64.1%**	**68.7%**

SS preserves accuracy better than WA at this operating point, with similar estimated energy savings.

SS: pruning ratio and observation-time trade-off

		1 hidden layer		2 hidden layers	
Observe	Pruned	Acc.	ΔE	Acc.	ΔE
4/20	50%	95.04	33.9	95.26	36.3
4/20	75%	95.00	50.1	95.26	54.6
8/20	90%	95.28	48.1	96.08	51.5
12/20	90%	95.52	32.1	96.16	34.3

Accuracy: All readout (%). ΔE: estimated neuron-energy reduction (%).

CIFAR-10: 90% middle-channel pruning

Observe	WA All	WA P2	SS All	SS P2
Baseline	92.90	–	92.90	–
1/6	20.50	20.48	20.28	20.27
3/6	67.00	67.07	66.50	66.88
5/6	**92.83**	92.32	**92.70**	92.35

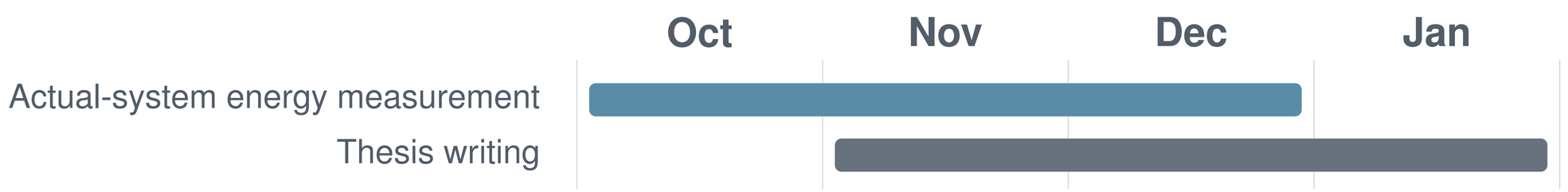
All/P2: accuracy (%) over all timesteps / Phase 2 only. Late pruning preserves accuracy, but affects only the final timestep. Under uniform, linear scaling, $0.90 \times 1/6 = 15\%$ of eligible-layer work is removed before overhead; this is an illustration, not measured energy.

MNIST STDP: 99% excitatory-neuron pruning

Observe	WA All	WA P2	SS All	SS P2
Baseline	92.15	92.14	92.15	92.14
25/250	64.73	64.75	64.74	64.76
50/250	91.01	90.95	91.01	90.97
75/250	**91.87**	91.82	**91.93**	91.90

At 75/250, accuracy stays within 0.3 percentage points of baseline while disabling 99% of excitatory neurons for the remaining 70% of inference. Early STDP pruning is limited by zero-score ties: at 25 timesteps, at least 475 of 500 neurons have not fired.

SCHEDULE



CONCLUSION

- Early spikes support per-sample structured pruning across FC, CNN, and STDP-trained SNNs without gradient updates during inference.
- Observation time controls the accuracy–savings trade-off. MLP neuron-energy savings are estimated; actual-system energy remains to be measured.

SELECTED REFERENCES

[1] Y. Kim and P. Panda, “Revisiting Batch Normalization for Training Low-Latency Deep Spiking Neural Networks From Scratch,” *Frontiers in Neuroscience*, 15, 773954, 2021.
[2] P. U. Diehl and M. Cook, “Unsupervised learning of digit recognition using spike-timing-dependent plasticity,” *Frontiers in Computational Neuroscience*, 9, 99, 2015.
[3] H. Hazan et al., “BindsNET: A Machine Learning-Oriented Spiking Neural Networks Library in Python,” *Frontiers in Neuroinformatics*, 12, 89, 2018.